
Outreach Statement – Sean Patrick MacBride

I applied to this fellowship because I want a career built on the same premise as Leon Lederman's: spend as much energy pushing the boundaries of physics as I do communicating it to the public. My own path into physics began at a winter astronomy camp in Northeast Vermont, and every outreach activity I have taken on since - from public observatory nights at my undergraduate college to gravitational-wave talks at a city-wide museum night - has been an attempt to hand someone else the same spark. I will bring that fervor to Fermilab, using the Lederman Science Center, Fermilab's Office of Education and Public Engagement, and the wider Chicagoland science community as a platform to communicate and educate others, both children and adults, about general principles of physics, dark energy, and gravitational-waves.

Past Outreach, Teaching, and Mentoring Experience: Intro statement

As an undergraduate at Wheaton College (2016-2020), I led public telescope nights, helping visitors of all backgrounds find and understand objects in the night sky using small 6" telescopes. This was my first experience translating a technical instrument into a shared, human experience, and was a seminal event that solidified my interest and desire to reach the public using science. As a graduate student at University of Michigan-Ann Arbor, I worked with the *god damnit what was their name* to lead public demonstrations with elementary and middle school kids during Metro Detroit Youth Day, including magnetic induction and the classical rotational inertia experiment on an office chair. During the 2023 annular and 2024 total solar eclipses, I took advantage of the cosmic coincidence to lead a group of Vermont high-school students from my hometown through lectures and demonstrations about the physics of the eclipses, eclipse safety, and ultimately observations from the path of totality. In 2025, I was a guide for visiting parties at Rubin Observatory as we commissioned the LSST Camera, translating the same instrument and technical knowledge to audiences ranging from political dignitaries, to NSF and DOE review teams, and friends and family visitors. In addition to providing tours, I gave informational interviews to the BBC, the Wall Street Journal, and Science News about LSST-Camera, and commissioning the observatory. Most recently, in September 2026, I led *Ask a Scientist* sessions on cosmology and gravitational waves at the University of Zurich's Science Pavilion for the City of Zurich's Long Night of Museums, an annual event in which the city's science, art, and culture museums open their doors free of charge.

Some intro for teaching Mention TAing at Wheaton That instinct carried into the classroom: I have been the lead or sole instructor for large introductory physics courses at University of Michigan-Ann Arbor and, currently, the University of Zurich. *Mention the Masters level teaching at UZH, and mention the difference between the teaching styles*

Some intro for Mentoring After a semester of teaching at University of Michigan-Ann Arbor, I recognized the level of stress and pressure many undergraduate students were feeling relative to an intense early-undergraduate curriculum. To address these concerns, I designed a public-facing art installation in the Michigan physics department, documenting the struggles and eventual successes of the department's own graduates and faculty members, aimed at demystifying what a career in science actually looks like - not just success, but failure and persistence to overcome a variety of obstacles.

Since 2023 I have mentored six undergraduate and master's students at University of Michigan-Ann Arbor and University of Zurich, guiding each through a first research project, from formulating the reaserch question to formalizing and presenting the result. Three are now pursuing physics PhDs of their own (at University of Zurich, the Paul Scherrer Institute, and CU Boulder) and one is a controls engineer at Argonne National Laboratory. I believe that one-on-one mentoring with passionate students provides the opportunity for a deeper appreciation of science, tailored to their own skill-set and passions.

My Plan for Outreach as a Lederman Fellow: As a Lederman Fellow, I want to pair frontier research, not only my own, with public science communication, building on the Dark Matter Days events, science-trivia nights, and QICK tutorials for the quantum program. I will extend this model to gravitational-wave and multi-messenger cosmology, a topic that easily captivates, but can feel intimidating for many.

To augment Saturday Morning physics, I would propose and give talks on gravitational-wave discoveries and observational astrophysics. These presentations will increase in relevance as the LSST continues on its ten-year survey, and new discoveries are made from the wealth of data. In addition to presenting about observational astronomy, I want to play a key role in helping the wider public become more familiar with gravitational waves, especially as new discoveries are made, including a future bright siren.

Working with the Fermilab Education and Public Outreach program, I will work to bring 'Ask a Scientist' sessions to the Lederman Science center. In these sessions, I would use them to connect to my own research expertise, but also incorporate other experts at Fermilab to emphasize the strong neutrino and dark-matter programs. The organization can follow a similar event structure to the Dark Matter Days, but instead of showcasing different experiments and concepts, have different timeslots during the 'Ask a Scientist' day to engage with Fermilab scientists that work on the energy, quantum, and cosmic frontiers.

I am eager to bring my outreach activities to the wider Chicagoland area, expanding beyond the Fermilab grounds. - Museum of Science and Industry - Adler Planetarium - UoC 'Astronomy Conversations' - Astronomy on tap?

In addition to recurring outreach activities in Chicago and at Fermilab, I would like to leave a legacy at the Lederman Education center by building a gravitational wave module using lycra and two rotating steel ball-bearings to help visualize the propagation of gravitational waves. This kind of visualization will help interested members of the public experience gravitational waves using one of the best senses to connect with a scientific concept - their eyes. While this module could serve as a permanent fixture in the Lederman Science center, I will work to build a similar module that can be used in QuarkNet teacher workshops, helping to add a display for another fundamental force.

Building on these outreach programs, I will serve as an active and engaged mentor for undergraduate and high school students. In my career, active mentors have had an immensely positive influence on my own trajectory. To pay it forward, I will volunteer as a judge and project mentor for the Illinois Junior Academy of Science's regional expositions and the CPS Student STEM Exhibition & Symposium, both of which actively recruit scientists with physics and astronomy backgrounds from the Chicagoland area. In addition to mentoring groups of junior students, I will take on students through Fermilab's TARGET high-school internship program, and the SULI and CCI undergraduate internship program, extending the same one-on-one research mentorship that has guided myself to a variety of rewarding opportunities in science to the next generation of fervent scientists.

Taken together, these are not activities I would bolt onto the Lederman Fellowship as an afterthought – they are a continuation of what I have already been doing at every stage of my career, now with the platform, the institutional history, and the surrounding Chicagoland science ecosystem to turn a plan into reality. I would be proud to carry that piece of Leon Lederman's legacy forward as a fellow.